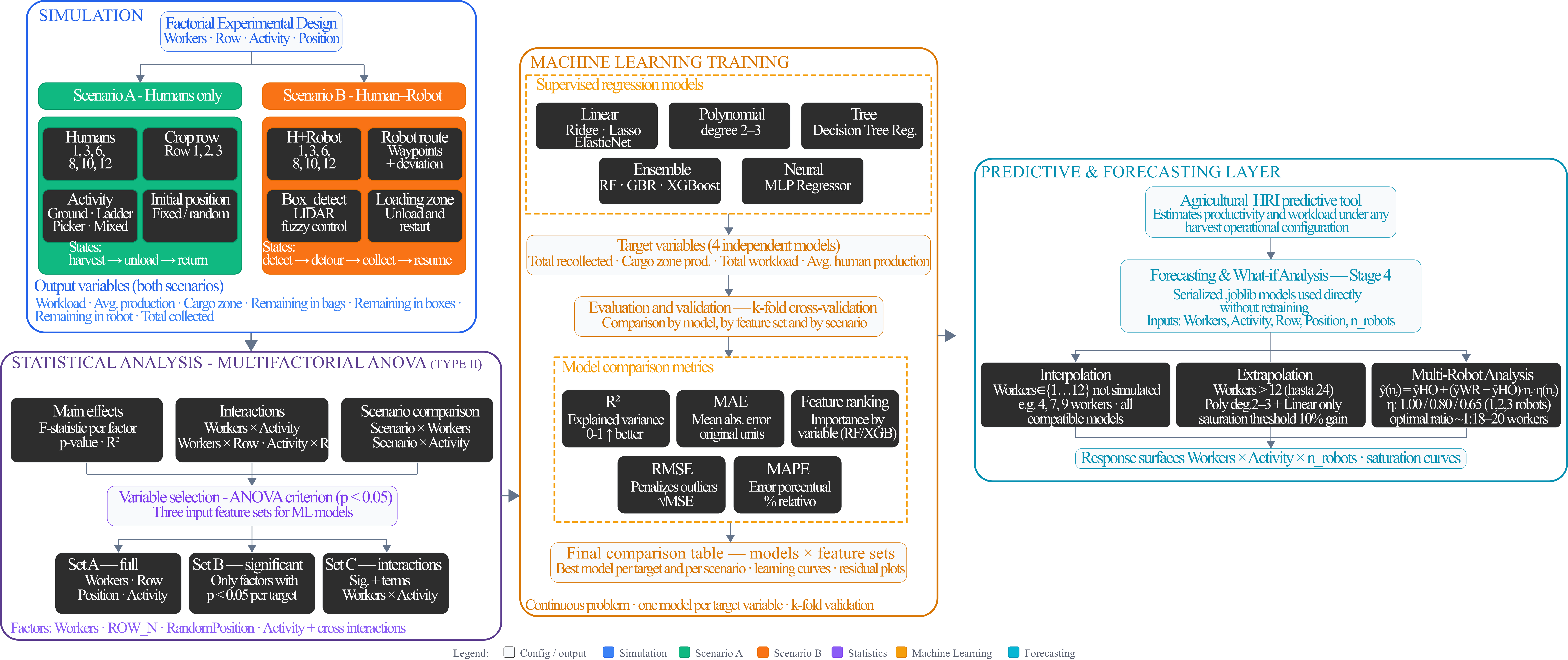
SIMULATION
Factorial Experimental Design
Workers · Row · Activity · Position
Scenario A - Humans only
Humans
1, 3, 6,
8, 10, 12
Crop row
Row 1, 2, 3
Activity
Ground · Ladder
Picker · Mixed
Initial position
Fixed / random
States:
harvest → unload → return
Scenario B - Human–Robot
H+Robot
1, 3, 6,
8, 10, 12
Robot route
Waypoints
+ deviation
Box detect
LIDAR
fuzzy control
Loading zone
Unload and
restart
States:
detect → detour → collect → resume
Output variables (both scenarios)
Workload · Avg. production · Cargo zone · Remaining in bags · Remaining in boxes ·
Remaining in robot · Total collected
STATISTICAL ANALYSIS - MULTIFACTORIAL ANOVA (TYPE II)
Main effects
F-statistic per factor
p-value · R²
Interactions
Workers × Activity
Workers × Row · Activity × R
Scenario comparison
Scenario × Workers
Scenario × Activity
Variable selection - ANOVA criterion (p < 0.05)
Three input feature sets for ML models
Set A — full
Workers · Row
Position · Activity
Set B — significant
Only factors with
p < 0.05 per target
Set C — interactions
Sig. + terms
Workers × Activity
Factors: Workers · ROW_N · RandomPosition · Activity + cross interactions
MACHINE LEARNING TRAINING
Supervised regression models
Linear
Ridge · Lasso
ElasticNet
Polynomial
degree 2–3
Tree
Decision Tree Reg.
Ensemble
RF · GBR · XGBoost
Neural
MLP Regressor
Target variables (4 independent models)
Total recollected · Cargo zone prod. · Total workload · Avg. human production
Evaluation and validation — k-fold cross-validation
Comparison by model, by feature set and by scenario
Model comparison metrics
R²
Explained variance
0-1 ↑ better
MAE
Mean abs. error
original units
Feature ranking
Importance by
variable (RF/XGB)
RMSE
Penalizes outliers
√MSE
MAPE
Error porcentual
% relativo
Final comparison table — models × feature sets
Best model per target and per scenario · learning curves · residual plots
Continuous problem · one model per target variable · k-fold validation
PREDICTIVE & FORECASTING LAYER
Agricultural HRI predictive tool
Estimates productivity and workload under any
harvest operational configuration
Forecasting & What-if Analysis — Stage 4
Serialized .joblib models used directly
without retraining
Inputs: Workers, Activity, Row, Position, n_robots
Interpolation
Workers∈{1…12} not simulated
e.g. 4, 7, 9 workers · all
compatible models
Extrapolation
Workers > 12 (hasta 24)
Poly deg.2–3 + Linear only
saturation threshold 10% gain
Multi-Robot Analysis
ŷ(nᵣ) = ŷHO + (ŷWR − ŷHO)·nᵣ·η(nᵣ)
η: 1.00 / 0.80 / 0.65 (1,2,3 robots)
optimal ratio ~1:18–20 workers
Response surfaces Workers × Activity × n_robots · saturation curves
Legend:
Config / output
Simulation
Scenario A
Scenario B
Statistics
Machine Learning
Forecasting